Unsupervised Deep Audio Denoising Algorithms via Hybrid Reinforcement Learning and Genetic Algorithm Meta-Learning

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Code (ReelSynth engine + DenoiseOpt meta-search): https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta

ABSTRACT

When a single-period wavetable (or any short cyclic buffer) is played with phase wrap, a sample discontinuity at the period boundary injects a broadband transient that repeats at the fundamental. We treat this cycle-local seam crackle as an audio denoising problem: seam-local operators (DualCosine crossfade, polish, FIR blends, optional tiny residual cells) are baked into the period, then judged by a prolonged residual $R \in [0,1]$ against an ideal multi-period tiling that withholds the open-wrap cliff (1 = best). We call the residual-scored hybrid search **DenoiseOpt**. An outer hybrid meta-search interleaves GA tournament updates with proximal policy optimization (PPO) architecture proposals, optional PBT exploit–mutate schedules, and MoE soft gates over heterogeneous bake cells. Candidates need not be paired with studio-clean waveforms for every trial. On a matched CUDA inference bench, a compact favorite reaches $R \approx 0.991$ versus DualCosine $R \approx 0.820$ ($\Delta R \approx +0.171$). A live GPU hybrid campaign has passed 5,000 clean iterations with champion $R \approx 0.9909$ against DualCosine ≈ 0.8166 on the runner geometry. Declared long-horizon mean-R tables remain open until the full budget finishes.

Keywords unsupervised learning · audio denoising · DenoiseOpt · neural architecture search · reinforcement learning · genetic algorithms · wavetable synthesis · wrap discontinuity · meta-learning

1 Introduction

Wavetable oscillators store one period of a waveform and advance a read pointer modulo the table length. If $x[0] \neq x[L-1]$, the wrap is a hard discontinuity. Under cyclic playback it becomes a periodic click whose spectrum is broadband and locked to the fundamental. The same geometry appears whenever a short loop buffer is tiled for listening or for objective evaluation. We call the repair task *cycle-local seam restoration*: edit a neighborhood of the wrap (and optionally a lightweight residual cell over the period) so that multi-period playback matches an ideal reference that never opens the wrap cliff. That outer agreement is what we score as audio denoising for this periodic artifact class.

Classical virtual-analog practice already supplies seam-local tools. Alias-free BLIT/BLEP-style synthesis [1], Poly-BLEP / fractional-delay antialiasing [3], and BLAMP corner corrections [2] reduce wrap and geometric discontinuity energy in-band. Those operators act locally. They leave open which bake parameters and which residual cell, if any, minimize prolonged tiled error against a no-cliff ideal. Label-free restoration work such as Noise2Noise shows that useful reconstructors can be ranked without paired clean targets when the evaluation geometry is explicit [4]. Speech-domain Noise2Noise denoisers extend that idea to audio waveforms [5]. Mixture-invariant unsupervised separation likewise avoids paired clean stems for ranking [29]. On the search side, NAS surveys [16], RL controllers [17, 18], aging evolution [20, 21], population-based hyperparameter schedules [22], CMA-ES tutorials [24], and evolution-guided policy gradients [23] motivate a hybrid outer loop: GA for population diversity, PPO for discrete architecture edits [19], and PBT-style exploit–mutate on elites. Platform-aware and progressive NAS add compute-aware search priors [27, 28]. MoE soft gates suggest routing among heterogeneous tiny cells under one residual score [25]. Waveform separation nets (Wave-U-Net, Conv-TasNet, dual-path RNN, SepFormer) and audio DL surveys inform the searchable cell family at bake-scale width [7–9, 11, 12, 30, 31], with U-Net / ResNet / attention primitives as block ancestors [10, 13–15].

The contribution is a meta-learning protocol for this periodic instance. An outer RL+GA(+PBT) loop proposes denoising operator graphs and continuous bake parameters. Each candidate is scored by a prolonged residual $R \in [0,1]$ (1 = best) between engine-baked multi-period audio and a procedural ideal tiling. Paired studio-clean wavetables are not required for every trial. Differentiable DSP modules [6] place DualCosine / FIR / MLP seam ops in an interpretable stack. Bayesian HPO [26] informs prior families that compete under the same R. A CUDA overnight campaign (PPO+GA+PBT+NAS, depth and MoE biases) has passed a 5,000-iteration clean gate with champion $R \approx 0.9909$ versus DualCosine ≈ 0.8166 on the runner geometry. Held-out matrices and declared long-horizon mean-R remain deferred until the full budget completes.

Contributions.

1. **Problem framing.** Cast wavetable wrap discontinuity / cycle-local seam restoration as audio denoising under prolonged tiled evaluation, separating seam-local diagnostics from residual R.

2. **Hybrid DenoiseOpt outer loop.** Specify RL+GA(+PBT) search over discrete operator graphs and continuous bake parameters, with named branches (GA crossover, PPO mutation, PBT exploit, MoE soft gates).

3. **OA literature protocol.** Separate used versus screened prior work for audio denoise cells and RL–evolutionary hybrids, grounded on downloaded OA PDFs only.

4. **Open evaluation path.** Release companion code for the synthesizer engine and meta search, with frozen 5k-gate figures and matched classical-vs-AI benches.

Section 2 organizes related work by mechanism. Sections 3–4 outline the residual objective and search protocol. Results report the 5k-gate freeze and inference benches.

2 Related Work

We organize prior work by mechanism. **Used** lines are methodological or comparative anchors for residual-scored RL+GA meta-search. **Screened** lines were inspected as contrast but are not treated as dependencies. Every citation below has a downloaded OA PDF in the companion meta repository. Paraphrases prefer attached abstracts and OA PDF snippets over title-only citation.

2.1 Discontinuity control and modular audio DSP

Alias-free digital synthesis of classic analog waveforms (BLIT/BLEP lineage) states the wrap discontinuity problem for trivial oscillators [1]. PolyBLEP-style fractional-delay corrections give practical antialiasing oscillators for subtractive synthesis [3]. BLAMP corner rounding treats discontinuities in the first derivative (triangular edges, clipping, rectification) without heavy oversampling [2]. DDSP reframes DSP modules as composable blocks inside learning pipelines [6]. We use that framing for small bake/FIR/MLP seam cells. We do not train end-to-end neural vocoders for every meta trial. These works are **used** as domain anchors for the operator vocabulary and for why prolonged cyclic playback is the outer judge.

2.2 Label-free restoration

Noise2Noise shows that restoration mappings can be learned from corrupted observations alone by exploiting structure in the corruption process [4]. Speech Noise2Noise denoisers apply the same idea to waveforms without clean paired audio [5]. Mixture invariant training ranks unsupervised separation systems without stem labels [29]. That philosophy motivates ranking denoise candidates without paired studio-clean wavetable cycles. The residual compares an engine-baked tiling to a procedural ideal that withholds open-wrap cliffs. We **use** this family as conceptual justification for unsupervised residual ranking. We do not retrain a Noise2Noise CNN on images.

2.3 Waveform denoising and separation cells

Deep learning for audio surveys the move from hand-designed features to learned time/frequency models [7]. Wave-U-Net performs end-to-end source separation with multi-scale 1-D U-Nets [8]. Improved Wave-U-Net speech enhancement and singing-voice U-Nets popularized encoder–decoder skips on audio [9, 10]. The original U-Net, ResNet residuals, and Transformer attention supply the block primitives we shrink to bake width [13–15]. Conv-TasNet and dual-path RNN demonstrated strong time-domain separation with temporal convolutions and long-context paths [11, 12]. SepFormer and audio spectrogram transformers push attention-heavy audio stacks further [30, 31]. These lines are **used** as design priors for tiny searchable cells (shallow U-Net / dilated / dual-path / attention blocks under a strict per-trial bake budget). They are not frozen full-scale separators inside every meta trial.

2.4 NAS and RL controllers

Elsken et al. taxonomize NAS by search space, strategy, and evaluation cost [16]. Zoph and Le couple RL controllers to architecture search [17]. ENAS adds parameter sharing for efficient discrete search [18]. Progressive and platform-aware NAS add staged evaluation and latency-aware rewards [27, 28]. We **use** this family as the ancestor of RL proposing architecture edits (add/remove/mutate ops, toggle residual-primary, mutate MLP/FIR), scored with prolonged residual. PPO supplies the clipped-surrogate policy update when the overnight loop logs PPO mutation branches [19].

2.5 Evolutionary NAS and population schedules

Large-scale and regularized / aging evolution showed that evolutionary NAS can discover competitive image classifiers when population turnover is carefully regularized [20, 21]. Population Based Training jointly adapts weights and hyperparameters in an asynchronous population [22]. Practical Bayesian optimization [26] and CMA-ES tutorials [24] inform local Bayesian and continuous evolutionary prior families. These are **used** for GA tournament/crossover branches, PBT-style exploit–mutate, and literature-combo priors, implemented at synthesizer-trial scale.

2.6 Evolutionary reinforcement learning hybrids

Khadka and Tumer's evolution-guided policy gradient (ERL) interleaves an evolutionary population with an RL actor to mitigate sparse reward and brittle hyperparameters in deep RL [23]. We **use** ERL as the spirit of interleaving GA population steps with PPO updates at tiny population sizes. We do not claim a full ERL robotics stack.

2.7 Mixture-of-experts soft gating

Sparsely gated MoE layers increase capacity via conditional computation [25]. We **use** MoE as motivation for soft gates over heterogeneous seam blocks (e.g., U-Net vs dilated vs attention cells) under residual scoring, at bake-scale widths.

2.8 Used versus screened

Used (methodological or comparative anchors).

- Discontinuity / VA lineage: [1–3].
- Unsupervised restoration philosophy: [4, 5, 29].
- Modular / differentiable audio DSP context: [6].
- Audio DL and compact waveform cells: [7–15, 30, 31].
- NAS / RL controllers / PPO: [16–19, 27, 28].
- Evolutionary NAS and population HPO: [20–22, 24, 26].
- ERL hybrid outer loop: [23].
- MoE soft gating prior: [25].

Screened (relevant contrast, not dependencies).

- **MAML** [32]: bi-level / fast-adaptation framing. We do not differentiate through a deep task network. Inner fits are coordinate-style updates on bake parameters. Outer rank is residual R.
- **DARTS** [33]: continuous relaxation of architecture search. We keep discrete ops + small cells under residual scoring.
- **Demucs / Hybrid Demucs / realtime waveform enhancement** [34–36]: strong separators and enhancers, but full multi-scale / hybrid STFT stacks are too heavy per meta trial under overnight bake budgets.
- **DiffWave** [37]: diffusion vocoder. Screened as a generative sampling stack, not a seam bake operator.
- **SEGAN-style adversarial enhancement** [38]: optional tiny adversarial cues only. Full GAN training loops screened for per-trial cost.

Positioning. Relative to this map, the present work contributes (i) a prolonged residual objective for periodic seam denoising, (ii) an explicitly hybrid RL+GA(+PBT) meta-outer loop with honest branch naming, and (iii) a used/screened literature protocol tied to OA PDF downloads. We do not claim SOTA on general speech enhancement or music separation corpora. The primary claim is protocol-level: residual-scored meta-search for cycle-local audio denoise operators under a declared compute budget.

3 Methods

Residual score. Let y be the engine-baked cycle and $r^\star$ an ideal multi-period reference from the same seed with open-wrap cliffs withheld. With tiling length N (typically $N{=}16$),

$$R = \text{clamp}\left(1 - \frac{\text{rms}(y_{\text{tiled}} - r^\star_{\text{tiled}})}{\max(\text{rms}(r^\star_{\text{tiled}}), \varepsilon)}, 0, 1\right). \tag{1}$$

By construction $R \in [0,1]$ with 1 best. Soft shape gates (when enabled) down-weight mid-cycle damage relative to seam-local error.

Searchable operator. Candidates combine classical seam ops (DualCosine, polish, soft seam, FIR blends) with tiny residual cells (MLP/FIR/U-Net-lite/attention-lite) and optional MoE soft gates. Bake parameters θ live in a bounded box. Architecture strings are discrete graphs over those ops.

Hybrid outer loop. Each iteration may (i) run GA tournament selection with crossover/mutation, (ii) sample PPO actions that edit ops/hyperparameters, (iii) apply PBT-style exploit–mutate on elites, and (iv) evaluate literature-combo priors. Selection is always by residual R (and logged auxiliaries). The live overnight configuration is tagged `PPO+GA+PBT+NAS+depth+MoE` with population size 12, plateau adaptation every 1000 iterations, and CUDA scoring on an RTX 3090.

4 Experiments

Overnight hybrid campaign (5k clean gate). A CUDA search tagged `PPO+GA+PBT+NAS` (depth and MoE biases) runs with a large declared iteration budget (order 10^5). At the 5,000-iteration clean gate the live job reports champion residual $R \approx 0.99093$ against DualCosine baseline ≈ 0.81658 on the same runner geometry ($\Delta R \approx +0.174$). Branch-best residuals at the freeze: GA ≈ 0.99016, PBT ≈ 0.99012, PPO ≈ 0.98924, combo ≈ 0.98854, NAS ≈ 0.98677. Figures in Section 5 regenerate from `history.jsonl` at this gate. Declared long-horizon mean-R is *not* claimed here. The GPU job continues past the gate.

Inference and classical matched benches. High-R fitted cells ($R \geq 0.98$) are timed on CUDA (batch 64, prolonged residual). Ten non-learnable bake denoisers share the same residual protocol against the neural favorite. Family-conditional hardness uses the separate Rust procedural 100,000-cycle bench reported in Results.

Table 1: Top-5 distinct fitted architectures on the CUDA inference bench (batch 64). Favorite is latency-best inside the near-max-R band.

Tag	R_{saved}	R_{live}	ms/batch	params
`iter_000157`	0.9914	0.9913	7.18	104 484
`iter_000205`	0.9911	0.9907	7.29	70 562
`iter_000235*`	0.9910	0.9915	3.15	37 489
`iter_000397`	0.9910	0.9907	6.80	97 262
`iter_000229`	0.9909	0.9902	6.67	100 500

*Favorite (fastest in $R \geq R_{\max} - 5 \cdot 10^{-4}$ band).

5 Results

Family-conditional hardness (completed bench). On the fitted DenoiseOpt 100,000-cycle bench, denoise score D is lowest for `nonlinear` ($D \approx 0.39$) and `combo` ($D \approx 0.40$), then `triple_mix` / `extreme_overlay` ($D \approx 0.51$–0.52), while `harmonic_fft` and `am_fm` sit near $D \approx 0.69$–0.70. Lit-combo champion family stress (prolonged residual) similarly ranks `open_wrap_bias` ($R \approx 0.83$) and `extreme_overlay` / `combo` ($R \approx 0.88$–0.89) below `harmonic_fft` ($R \approx 0.95$). Hard sounds recur.

Overnight hybrid (5k-gate freeze). At $\geq 5{,}000$ clean iterations the champion residual is $R \approx 0.99093$ with DualCosine ≈ 0.81658 on the overnight runner ($\Delta R \approx +0.174$). Monitoring plots below are regenerated from that freeze. They are *not* a finished declared-budget mean-R claim.

Inference time at matched score. Across diverse high-R fitted cells ($R \geq 0.98$), we timed CUDA forward passes (batch 64, prolonged residual). Within a tight band of the best residual ($R \geq R_{\max} - 5 \cdot 10^{-4}$), the **favorite** is the lowest-latency model: `champion_iter_000235` with $R \approx 0.9910$, ≈ 3.15 ms/batch, ≈ 37k parameters (versus a higher-capacity max-R cell at $R \approx 0.9914$ but ≈ 7.18 ms/batch / ≈ 104k params). Selection rule: maximize R, then minimize latency inside the band.

5.1 Top-5 architectures

Table 1 ranks five distinct fitted bake graphs from the CUDA inference bench (`inference_bench.json`, batch 64), preferring residual R first and latency on ties, and skipping near-duplicate copies of the same topology signature.

Classical vs AI (same residual protocol). On the overnight runner geometry (CUDA, batch 64, prolonged residual R), ten non-learnable bake denoisers were timed against the neural favorite above. DualCosine reproduces the paper baseline at $R \approx 0.820$ (≈ 1.19 ms/batch), consistent with the overnight monitor DualCosine ≈ 0.817. The best *active* classical method is a seam-local 3-tap FIR (`seam_fir3`) at $R \approx 0.963$ / ≈ 0.60 ms/batch. Identity (no-op) scores $R \approx 0.974$ and is reported only as a ceiling. The neural favorite reaches $R \approx 0.991$ at ≈ 2.47 ms/batch (≈ 37k params): $\Delta R \approx +0.028$ vs `seam_fir3` and $\Delta R \approx +0.171$ vs DualCosine, at roughly

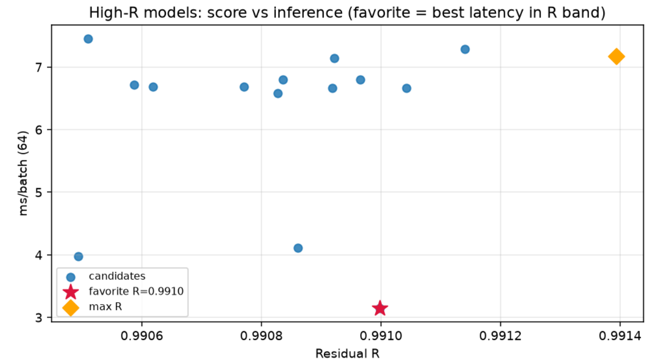

Figure 1: High-R architectures: residual versus CUDA inference latency (ms/batch). Star: favorite (fastest in the near-max-R band).

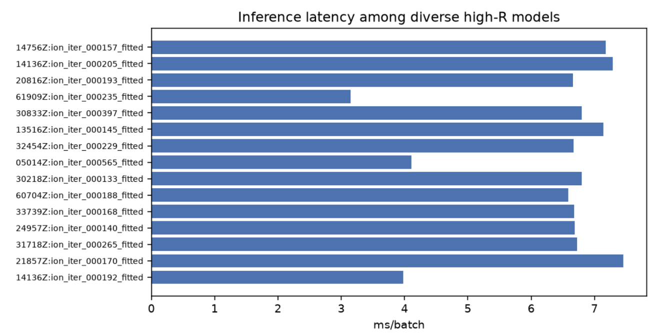

Figure 2: Inference latency bars for diverse high-R fitted models.

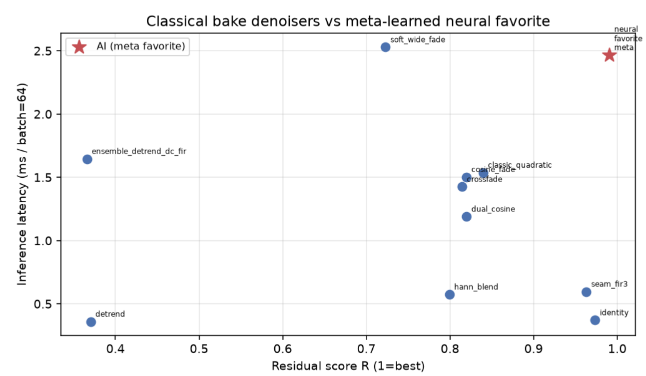

Figure 3: Classical bake denoisers vs neural favorite: residual R versus CUDA latency (ms/batch). Red star: AI favorite.

4.1× / 2.1× the latency of those classical baselines respectively.

6 Discussion

Where residual mass remains. Across the 100,000-cycle procedural bench and lit-combo family stress tests, wrap error is not uniform. The same generator families repeatedly dominate the residual budget: `nonlinear`, `combo`, `triple_mix`, and `extreme_overlay` show the weakest

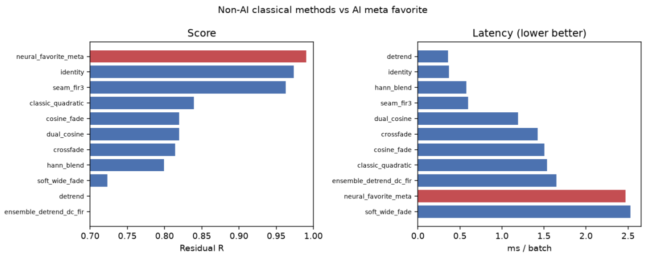

Figure 4: Classical vs AI ranking on residual (left) and latency (right). AI in red. DualCosine ≈ 0.820 on this protocol.

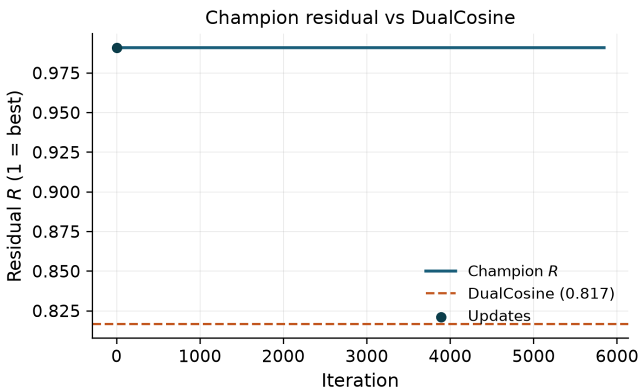

Figure 5: Champion residual R versus search iteration at the 5k-gate freeze, with DualCosine baseline.

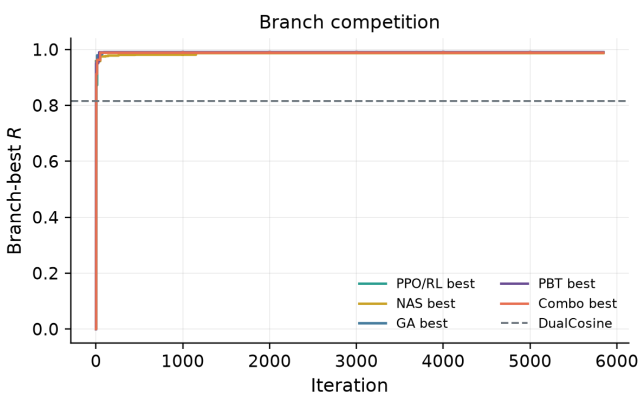

Figure 6: Per-branch best residual trajectories (PPO/RL, GA, PBT, NAS, combo) at the 5k-gate freeze.

denoise / residual scores, while `harmonic_fft` and `am_fm` are comparatively easy. `open_wrap_bias` can look strong on wrap-energy proxies yet still lag on prolonged residual. Cliff reduction and ideal-matched tiling are related objectives with different rankings.

Implication for this paper. Mean R (and mean DualCosine gaps) therefore compress a structured difficulty map. Reporting only the overnight mean on synthetic sine cliffs understates how much of the remaining $\sim$1% sits in a few hard families. That structure is a first-class empirical claim of the present work, independent of whether the GPU hybrid later nudges mean R from ≈ 0.99 toward 0.999.

Hybrid search vs. plateau. Overnight traces at the 5k gate show early high champions then long plateaus near $R \approx 0.99$. Plateau adaptation (deeper cells, denser MoE graphs, higher GA/PBT weight) is the operational response inside one campaign. It does not by itself rebalance family-conditional risk. A natural next paper is family-conditional and cliff-conditional meta-learning: train or select denoisers per family, or condition the outer loop on measured wrap statistics, and publish per-family residual curves beside the scalar champion.

Hybrid branches. At the 5k freeze, GA and PBT lead branch-best residuals, with PPO close behind and NAS trailing on this seed. Final win-rates after the declared GPU budget remain deferred.

7 Tooling and AI use

This section discloses research and writing tools, including AI systems, with scope and intent.

Literature discovery. Open literature was gathered with the Klaut Research Gateway (local OpenAlex / Crossref / arXiv providers) under queries spanning virtual-analog anti-aliasing, unsupervised restoration, HPO/PBT, RL–GA hybrids for NAS, and audio denoise architectures. Harvest JSONs live in the companion meta repository. OA PDFs were downloaded for every bibliography entry (`artifacts/literature_oa/pdfs/`). Non-OA items were dropped or replaced.

Paper drafting. Section planning, subsection drafts, and revision passes used Klaut `research_paper_*` tools with local Ollama (`qwen3.5:9b`) when available, under an explicit scientific workflow (plan → write → revise → export) into an arXiv-style twocolumn template. AI drafts were treated as scaffolding. Claims, numbers, and citations were checked against harvest records, PDF analyses, and measured artifacts. Final prose was edited for DSP terminology and to remove generic filler.

Why AI was used. (1) Scale literature triage across DSP and AutoML without pretending every high-cite hit is in-domain. (2) Accelerate IMRaD scaffolding so human effort focuses on metric honesty (residual vs DualCosine, family hardness, gate vs final overnight claims). (3) Keep an auditable tool chain (plans, PDF snippet analyses, OA flags) over opaque chat-only writing.

Where AI was *not* used. Model fitting, residual scoring, DualCosine baselines, lit-combo / overnight GPU search, and inference-time benchmarks are conventional code (Rust / PyTorch) with deterministic logs and JSON artifacts. AI did not invent experimental numbers.

Other tools. Rust/`cargo` for the procedural bench and meta binaries. PyTorch CUDA for overnight hybrid search.

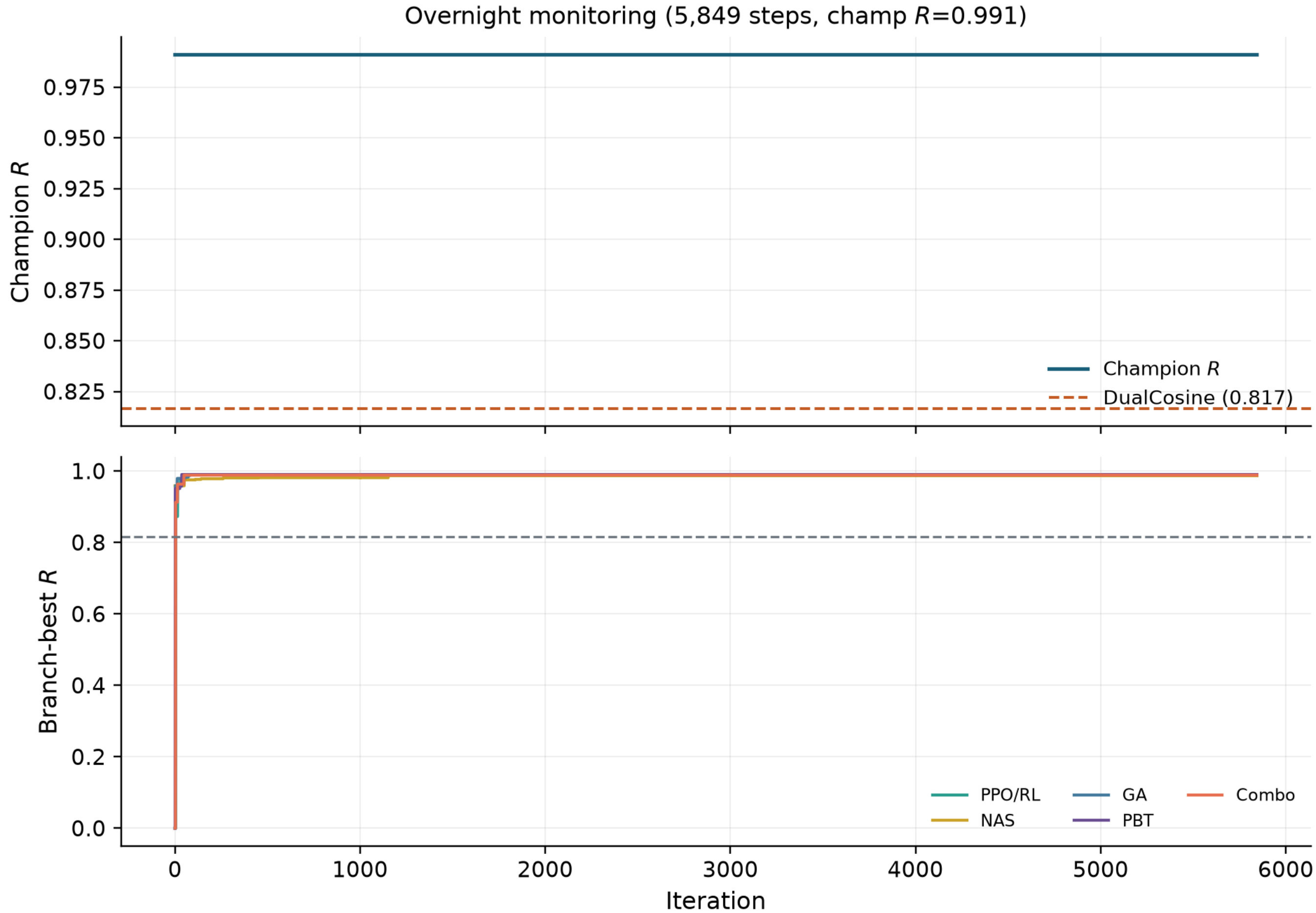

Figure 7: Full-width overnight monitoring panel at the 5k-gate freeze: champion R (top) and branch-best trajectories (bottom).

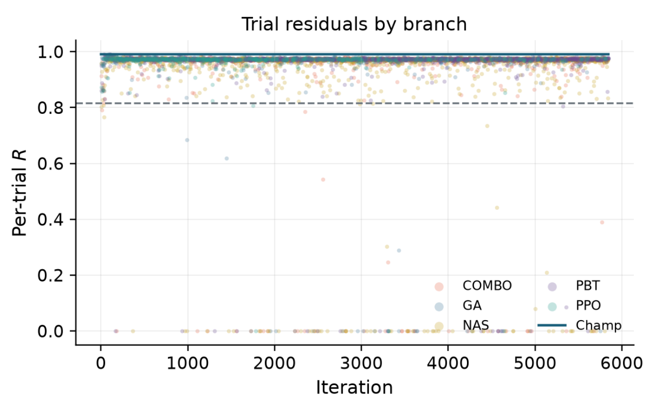

Figure 8: Per-trial residual scatter by search branch, with champion overlay (5k-gate freeze).

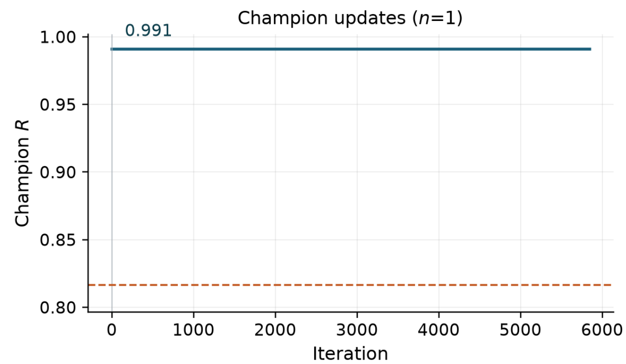

Figure 9: Champion update timeline. Sparse R labels mark first, peak-mid, and latest improvements to stay legible in twocolumn.

matplotlib for figures. MiKTeX `pdflatex` for PDF builds. GitHub for versioned papers and artifacts.

8 Limitations

- Residual uses a procedural ideal (no-cliff sibling), not a perceptual MOS or listening panel.
- Bake metrics are not a substitute for formal audio quality tests.
- The overnight GPU loop currently scores random sine+harmonic seam cliffs. Family-conditional hard-

ness is measured on the separate Rust procedural bench. Aligning both evaluators is left to follow-up work.

- Per-trial architecture width is capped for overnight throughput. Full Demucs-/diffusion-scale stacks were screened out.
- The 5k-gate freeze reports measured champion and branch residuals. Declared long-horizon mean-R still requires the completed budget and frozen seeds.
- Classical VA/BLEP anchors use OA author or proceedings PDFs (`stilson1996`, `nam2009polyblep`, `esqueda2016blamp`).

9 Outlook: follow-up publication

A self-contained follow-up can treat *which sounds fail* as the primary object of study: stratified residual heatmaps over the ten procedural families, cliff-size conditioning, and family-specialist or mixture-of-experts denoisers selected by the same RL–GA outer loop. That paper would cite the present mean-R hybrid protocol as the baseline and ask whether beating $R > 0.999$ is mainly an architecture problem or a coverage problem over hard families.

10 Conclusion

We framed wavetable wrap discontinuity as an audio denoising instance and specified DenoiseOpt, a hybrid RL+GA meta-learning outer loop scored by prolonged residual R. Related work separates used anchors (Noise2Noise / speech N2N, PBT, aging evolution, ERL, ENAS/PPO, Wave-U-Net/Conv-TasNet, MoE, BLIT/BLAMP/DPW, DDSP) from screened contrasts (MAML, DARTS, Demucs, DiffWave, full GAN loops), with OA PDFs for every cite. At the 5,000-iteration clean gate the overnight champion reaches $R \approx 0.9909$ versus DualCosine ≈ 0.8166. A compact inference favorite reaches $R \approx 0.991$ versus DualCosine $R \approx 0.820$ on a matched CUDA bench. Companion code lives at https://github.com/reeldemo/reelsynth and https://github.com/reeldemo/denoise-opt-meta. Declared long-horizon mean-R awaits completion of the ongoing GPU campaign.

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